

The Texas Interconnect: Renewables and Extreme Weather Conditions

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Background

ERCOT

The Electric Reliability Council of Texas (ERCOT) manages the Texas Interconnect, an alternating current (AC) power grid that covers about 90% of the state. ERCOT delivers electricity to 26 million people, generating 82,000 megawatts and connecting more than 46,500 miles of transmission. It is the country's first independent system operator (ISO) and is subject to oversight by the Public Utility Commission of Texas and the Federal Energy Regulatory Commission [1].

Winterization Challenges: Storm of 2021

In February 2021, extremely cold weather resulted in both high electrical demand as well as reduced generation capacity due to a lack of sufficient winterization, causing major blackouts across the state. Between February 15-27, more than 4.5 million homes and businesses lost power, some for many days [2]. This event was not the first of its kind in Texas. In 2011, a major winter storm caused 3.2 million Texan homes to lose power. At the time, ERCOT failed to change the winterization standards for its electricity generators, leaving the state vulnerable to a similarly damaging winter storm in 2021 [1].

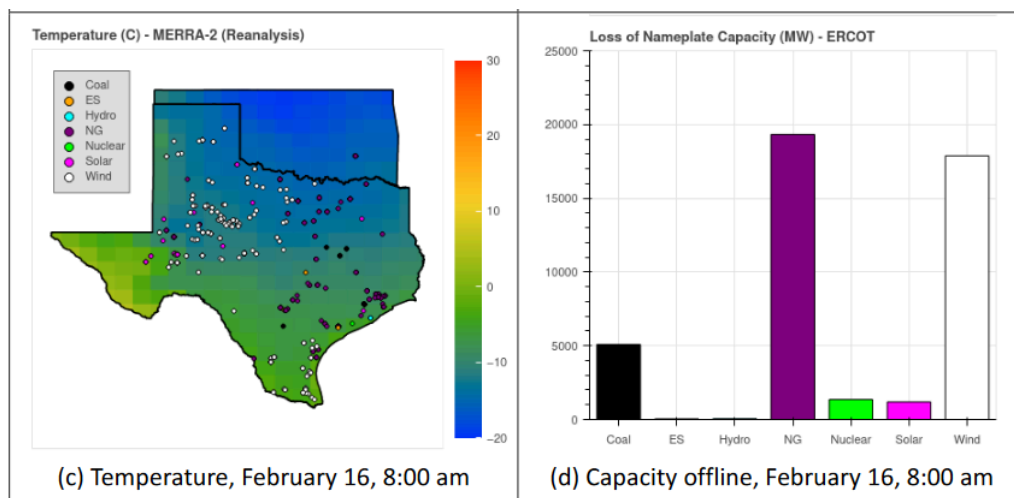


Figure 1: Temperature across Texas and reported loss of capacity by ERCOT between Feb. 15, 2021, 1:45 am and Feb. 16, 2021, 8:00 am [2]

However, the 10% of Texas counties not powered by ERCOT invested millions in winterization. This strategy protected their counties from suffering the same fate as those covered by ERCOT. For example, of those powered by El Paso Electric, an independent energy provider, only 3,000 out of around 440,000 homes lost power for 5 minutes during the 2021 storm [1]. After this storm, ERCOT received pressure to mandate winterization developments. ERCOT initially attributed blackouts to the wind and solar generation systems, when, in reality, it was in large part due to the management of the demand and transmission by ERCOT and freezing of natural

gas sources [3]. The storm raised many questions about how to prevent similar disasters in the future among Texans and energy researchers alike [2], such as:

- Could the risks of blackouts under extreme weather conditions be better estimated and mitigated?
- What (if any) changes should be made to reserves, storage capabilities, or other mitigation strategies?
- Is there an over-reliance on one type of generation? What should the mix look like to ensure reliability?
- How much winterization is needed and where should these efforts be prioritized?

Generation Mix

The events of February 2021 inspired a surge of investments in renewable energy generation capacity within the state. In 2021, Texas led the U.S. in new renewable energy projects by installing 7,352 MW of renewable energy sources. The state has proposed 106 GW of solar power projects [4]. ERCOT is taking advantage of Texas's area and climate conditions to scale up wind generation. Currently, 22.5% of the electric grid is powered by wind and proposed offshore wind projects are expected to power around 2.3 million homes [5].

Prior to the energy crisis, Texas was not nearly as invested in renewable energy generation. Texas's generation mix from 2007 as recorded by ERCOT is almost half gas at 44.5%, coal at 37.4%, nuclear at 13.4%, wind at 2.9%, and hydro and others making up 3.3% [6]. As of 2022, the generation mix has shifted to gas-cc 35%, wind 26%, coal 16%, nuclear 10%, gas 7%, and solar 6% [7]. Integration of renewables has increased in the past 20 years as their costs have decreased. Due to the independence of the Texas Interconnect and the resulting lack of federal mandates, there are not clear system wide mandates for renewable energy related to carbon neutrality on the generation side. Further, the storage needed for scaling up of renewables is not available yet [4] .

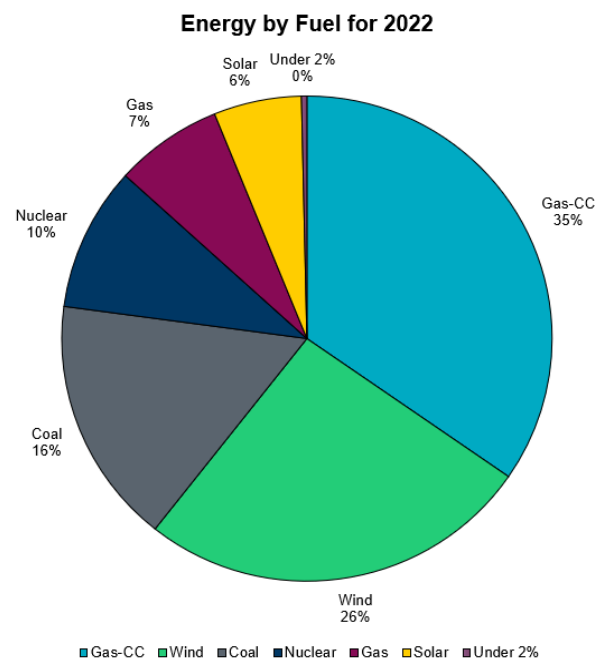


Figure 2: Texas energy generation mix in 2022 [7]

Texas is experiencing tremendous growth right now with a residential and corporate population boom and has a large capacity for wind and solar generation, but the lack of transmission and storage in the state means that the growth in renewable sources will be extremely slow. Consequently, Texas will require sustained investment in non-renewable sources if the state expects to meet the increasing demand [4].

Motivation and Scope

ERCOT came under intense scrutiny when the state-wide winter storm of early 2021 left residents without power and water for almost a week, exposing large-scale system unpreparedness, highlighting storage limitations and lack of winterization for renewable and non-renewable sources. The heightened attention that the storm placed on the pitfalls of the independent electricity grid managed by ERCOT in Texas provided our team a starting point for examining the ability for the state to focus on the growth and stability of solar and wind while also preparing for future weather events and managing increasing demand [2].

Using data from ERCOT on the fuel mix reports, demand data and forecasting reports, winterization reports, 2021 blackout reports, and energy storage project reports, our team is going to consider how to balance the goals of further renewable integration and improved weather resiliency in the context of increasing demand.

We will focus on how the increasing integration of renewables into the generation mix will impact the system's ability to handle expected vs. extreme system loads. Additionally, we will explore if and how winterization strategies need to be modified, especially in preparation for extreme system loads.

ERCOT's Analysis Methods

Economic Dispatch and Optimal Power Flow

ERCOT makes use of the Real-Time Security Constrained Economic Dispatch (RT-SCED) method to determine the optimal generation and dispatch of electricity of the power system in real time [8]. This method takes into account the system's constraints and limitations. The process of RT-SCED, an optimal power flow model, begins with the projection of the demand for electricity over a certain time period [9]. In the case of Texas, forecasting is updated every five minutes and predicts demand at least 15 minutes into the future. These projections are used to determine the amount of generation capacity that will be needed to meet the demand. ERCOT then calculates both binding Base Points and LMP's (the prices that will be used to dispatch generation resources) and a non-binding projection of the Base Points and LMP's. These forecasts are also updated every 5 minutes and project at least 15 minutes ahead [8].

Unit Commitment

ERCOT conducts at least one Hourly Reliability Unit Commitment (HRUC) and one Day-Ahead Reliability Unit Commitment (DRUC) before every operating hour. In addition, ERCOT can conduct a RUC at any point to analyze and resolve reliability issues. In evaluating RUC's, ERCOT makes use of Real-Time Co-optimization (RTC), which involves the interchangeable dispatch of energy and ancillary services in the Real-Time Market. Ancillary services support the power system's reliability and include frequency and energy reserve regulation as well as voltage control [10].

Methodology

Data

Table 1: Data obtained for the project

Type	Use
Weather Zones	ERCOT uses the weather zones to aggregate data, and in our project we apply state-wide data over weather zones [11].
Current Non-Renewable Generation Capacity	Used in all 6 cases to provide generation values from non-renewables. This generation is kept at the same value for the “expected” cases [12].
Current Renewable Generation Capacity	Used in the “current” cases to provide generation values from renewables [13].
Expected Renewable Generation Capacity	Used in “expected” cases to provide planned generation from renewables for up to 2032 [5].
Demands from Feb 2021, Feb 2022	The maximum demand on the load on the 2/12/21 was used as the “extreme” value for the cases, and in contrast, the maximum demand on 2/12/22 was used as the “base” value [13].
Generation Capacity offline during the winter storm	The data from this source was aggregated by the weather zone in order to create the percentages of available solar and wind in the extreme cases based on the capacities that were recorded offline [5].

Model

Our approach to understanding the effect of renewable integration and winterization on the Texas power grid was aided by the use of “*An open-source extendable model and corrective measure assessment of the 2021 Texas power outage*” developed by researchers at Texas A&M University and Seattle’s Breakthrough Energy Sciences [14]. The model built includes a realistic, baselines grid that was adapted from a test system and developed synthetically using data sets related to load and generation across the Texas Interconnect. The synthetic grid contains 606 generators, 1350 loads, and 3206 branches, and it has been calibrated using available open-source data [15]. We primarily used this model as a source of a base case that established the topography of the system, which we then modified for our various cases. We did not use the model’s built-in approach to representing the weather-induced capacity constraints and other features, rather representing these ourselves through the cases presented.

Approach

To process the data we first extracted internal matrices of a model mpc for an example matpower input done to theoretically represent the 2000 bus Texas system. We were then able to determine what columns within each matrix (representing branches, generators, and buses) needed to be modified. Our goal was to modify the mpc in six different ways to represent the following cases:

- Case 1: The **current generator mix**, with the current **standard load**, and **all generators available** for use.
- Case 2: The **current generator mix**, with the **extreme winter load**, and **generators limited** in availability.
- Case 3: The **future generator mix**, with the current **standard load**, and **all generators available** for use.
- Case 4: The **future generator mix**, with the **extreme winter load**, and **generators limited** in availability.
- Case 5: The **current generator mix**, with the **extreme winter load**, and **additional generator availability**.
- Case 6: The **future generator mix**, with the **extreme winter load**, and **additional generator availability**.

We expected failure for Cases 2 and 4 as Case 2 is represented by the snow storm failure in February and Case 4 represents how we would represent that with the future additional planned infrastructure and distribution. Each case was run with matpower using dcopf to process each of the six cases. Cases 2 and 4 were modified to get an understanding of what additional capacity would be needed to handle the extreme winter loads. We modified one of the existing generators to act as an “overflow” generator with an unlimited capacity and a high cost to ensure the model turned to the usage of that generator as a last resort. The overflow generator was also assumed to be a nonrenewable generator (gas, coal, or hydro) as we are measuring the potential limitations of the system when available renewable energy would be more limited. Case 5 and 6 are representations of the perfect situation under each of these cases where improved preparation allows for the extreme demand to be met.

Assumptions

In order to test our questions of the renewable integration and the impact on winterization and use the 2000-bus open-source model to carry this out, the following assumptions were made.

When representing capacity: for non-renewable sources, we made the simplifying assumption that current capacity would be spread equally across the weather zones and then divided equally among the non-renewable generators within that weather zone. For those, we used the current and expected wind/solar capacity by county, rolled up to weather zones, and then distributed equally among the appropriate generators within the weather zone. Additionally, between our current and future cases, we assumed non-renewable capacity did not change (i.e. that no non-renewable generators were permanently removed) and that, therefore, any new

renewable capacity would be additive. In a few cases, this required updating the model, as not every weather zone initially had at minimum one of each type of generator which caused issues in cases where we expected to allocate capacity to a certain generator type in a specific weather zone. As such, we updated the model in these specific instances to replace one non-renewable generator with the required renewable generator type and adjusted the capacities in the weather zone accordingly. This was likely required due to differences in the modeled topography and that expected based on our assumptions and the historical data.

When representing availability, we estimated the percent of capacity that would remain online during extreme weather conditions by weather zone and generator type based on data from the 2021 storm. We further assumed, for simplification, that all capacity was online during non-extreme weather conditions (i.e. not taking into consideration downtime/maintenance). We also assumed that the winterization ability and therefore generator availability percentage would hold as generator capacity increased, for example due to increased efficiency or improved practices. In other words, as additional generators came online between current and future scenarios, if 10% of the capacity was available in the current scenario, an equivalent 10% of the future capacity would be available in the future.

When calculating the load, we based the demand for the extreme case on the first high load day of the 2021 storm by weather zone and for the base case we used the demand of the same day in 2022, in an effort to minimize potential seasonal impact. In both scenarios, each weather zone's demand was distributed equally across buses within that zone.

Results

Current State: Normal and Extreme

In comparing the calculations for Case 1 and Case 2, the difference between the stressed and unstressed system is clear. In Case 1 (under normal conditions), the capacity at 72,716.3 MW is more than adequate to handle the 42,703.7 MW demand. By comparison, in Case 2, the system is subject to both a higher demand of 60,017.0 MW, for example due to increased need for heating in homes and businesses, but also finds its capacity greatly reduced due to the weather, down to 29,702.6 MW. As a result, the system would not be able to meet the demands during the extreme weather. As a note, our solution in Matpower did "show" a convergence, but note that it can be seen in the results that the load is far greater than the capacity, though the capacity did not correctly limit the generation. However, from these results, one can quickly identify that the system was under capacity by 30,314.4 MW. As an additional trial, we ran this case with an expensive generator with very high capacity (Case2- With Overflow) to highlight that this demand could be met with higher generation capacity.

Future State: Normal and Extreme

In Case 3 and Case 4, we made the same comparison as in Case 1 vs Case 2, but with the future generation mix. As we considered all new solar and wind capacity as additive, clearly the

system was capable of handling the base weather scenario (Demand: 42,703.7 MW vs. Capacity: 105,873.3 MW). What is interesting to note is that, even with the added overall capacity, the extreme weather limits that added capacity enough that the system still could not handle the extreme weather scenario (Demand: 60,017.0 MW vs. Capacity: 54,485.3 MW, for a difference of 5,531.7 MW). We also again demonstrate that with additional generation at high cost and large capacity, this demand could be met (Case4- With Overflow). Case 4 demonstrates the importance of winterization efforts as it highlights that even with significant added capacity, the reduced availability of un-winterized generators will still cause the system to fail to meet demand.

Improved Winterization: Current and Future State

In these cases, we wanted to demonstrate the value of improved winterization efforts. First, we ran scenarios to identify solutions where there was just enough improvement in generator 'availability' (i.e. winterized enough to remain at higher capacity during extreme weather) to meet the extreme weather scenario. We wanted to do so in order to demonstrate that in both scenarios, better winterization is needed to meet the demand and to highlight the necessary improvements. The range of percent improvements required varied by generator type and weather zone, which suggests a priority order these could be ranked. In Case 5, dramatic improvements were required (roughly 40-60%) on non-renewable generators in all weather zones and significant improvements (roughly 58%) to wind generators in zones more reliant on wind. The added capacity in the future scenario makes the required improvements much more reasonable: a range of roughly 10-15% on the non-renewable generators and targeted 5-6% improvements on wind generators. With these improvements, the capacity is able to meet the demand in both scenarios (Case 5: Demand: 60,017.0 MW vs Capacity: 60,180.6 MW, Case 6: Demand: 60,017 MW vs. Capacity: 60,023.1 MW).

Conclusions

Supply and Demand

The current capacity and generation mix meet base level demand under non-extreme conditions. That said, the current capacity and mix cannot handle extreme weather conditions. In this case, the supply/demand gap expands as demand increases and available capacity decreases. A demand increase can be attributed to an increased need for electricity (i.e. seasonal heating, cooling). Available capacity decreases due to weather-related outages impacting different generator types at different rates. This issue remains even as capacity is added through the addition and integration of renewable energy sources (i.e. wind and solar). In fact, if the assumption that current winterization percentages hold as the renewable capacity increases is not valid, the value of this added capacity could be even more limited in this case.

Winterization

Winterization efforts need to improve for each generator type in each weather zone, but the Far West, West, and North weather zones should be prioritized respectively based on our analysis.

In several feasible modeled solutions, these regions need the largest overall percent improvements to get to the required available capacity. While the results varied across several modeled solutions, priority within a given weather zone should typically be given to the non-renewable generators as they often need higher percent improvements than the renewable generators. In the case of the Texas Interconnect, the dependence on natural gas as its main non-renewable generator strengthens the need for winterization due to the freezing potential of the fuel. Additionally, as generation capacity is added via the renewable energy sources, ERCOT can reduce the improvement percentages necessary to reach the required available capacity to more achievable levels.

Potential Areas for Further Work

In order to strengthen our model, various changes can be enacted to decrease the assumptions made, specify the model, and better answer our questions relating to the power of renewable energy grid investment and prevention of future weather related outages. Specifically regarding non-renewable generation, assumptions on generator capacity could be replaced with accurate data. The accurate distribution of generating capacity across the weather zones can replace the current equal distribution, and the net change in generation capacity in each weather zone can replace the current state of our model of zero net change. Additionally, in our current model, we made the simplifying assumption that capacity and demand were generally equally distributed within weather zones across generators of the same type or buses, but further work could find more accurate results by using real data in place of this assumption. The estimation method our team used to relate the impact of weather in areas of the state and the generation capacity in those areas could be refined, such as by incorporating the true outages from the 2021 storm. In order to provide recommendations for winterization efforts that the state can take to strengthen the grid, the cost to perform this winterization can be investigated so that certain generation types or weather zones can be prioritized in preparation for future extreme weather events in both summer and winter.

Exhibits

Exhibit 1: Case 1: Current Gen Mix, Base Load, Full Availability

Zone	Base Cap Non	Base Cap Wind	Base Cap Solar	Base Avail Non	Base Avail Wind	Base Avail Solar	Total Non	Total Wind	Total Solar	Total Cap	Base Load
FAR WEST	4,003.91	5,143.90	5,270.70	100%	100%	100%	4,003.91	5,143.90	5,270.70	14,418.51	4,356.70
WEST	4,003.91	8,923.10	1,151.60	100%	100%	100%	4,003.91	8,923.10	1,151.60	14,078.61	1,240.30
EAST	4,003.91	0.00	309.80	100%	100%	100%	4,003.91	0.00	309.80	4,313.71	1,781.37
NORTH	4,003.91	6,266.90	1,343.80	100%	100%	100%	4,003.91	6,266.90	1,343.80	11,614.61	1,066.34
NORTH CENTRAL	4,003.91	2,247.70	540.90	100%	100%	100%	4,003.91	2,247.70	540.90	6,792.51	13,186.74
SOUTH CENTRAL	4,003.91	0.00	346.90	100%	100%	100%	4,003.91	0.00	346.90	4,350.81	6,622.34
SOUTH	4,003.91	7,726.30	608.21	100%	100%	100%	4,003.91	7,726.30	608.21	12,338.42	2,950.35
COAST	4,003.91	151.20	654.00	100%	100%	100%	4,003.91	151.20	654.00	4,809.11	11,499.55

Exhibit 2: Case 2: Current Gen Mix, Ext Load, Limited Availability

Zone	Base Cap Non	Base Cap Wind	Base Cap Solar	Ext Avail Non	Ext Avail Wind	Ext Avail Solar	Total Non	Total Wind	Total Solar	Total Cap	Ext Load
FAR WEST	4,003.91	5,143.90	5,270.70	25.07%	28%	84%	1,003.91	1,440.29	4,427.39	6,871.59	3,936.34
WEST	4,003.91	8,923.10	1,151.60	25.07%	28%	84%	1,003.91	2,498.47	967.34	4,469.72	1,768.64
EAST	4,003.91	0.00	309.80	25.07%	100%	90%	1,003.91	0.00	278.82	1,282.73	2,377.52
NORTH	4,003.91	6,266.90	1,343.80	25.07%	28%	84%	1,003.91	1,754.73	1,128.79	3,887.43	1,210.05
NORTH CENTRAL	4,003.91	2,247.70	540.90	25.07%	60%	87%	1,003.91	1,348.62	470.58	2,823.11	20,882.51
SOUTH CENTRAL	4,003.91	0.00	346.90	25.07%	100%	90%	1,003.91	0.00	312.21	1,316.12	10,458.69
SOUTH	4,003.91	7,726.30	608.21	25.07%	75%	90%	1,003.91	5,794.73	547.39	7,346.02	4,735.78
COAST	4,003.91	151.20	654.00	25.07%	75%	90%	1,003.91	113.40	588.60	1,705.91	14,647.48

Exhibit 3: Case 3: Future Gen Mix, Base Load, Full Availability

Zone	Fut Cap Non	Fut Cap Wind	Fut Cap Solar	Base Avail Non	Base Avail Wind	Base Avail Solar	Total Non	Total Wind	Total Solar	Total Cap	Base Load
FAR WEST	4,003.91	6,867.58	6,505.29	100%	100%	100%	4,003.91	6,867.58	6,505.29	17,376.78	4,356.70
WEST	4,003.91	11,361.10	3,595.21	100%	100%	100%	4,003.91	11,361.10	3,595.21	18,960.22	1,240.30
EAST	4,003.91	0.00	1,930.23	100%	100%	100%	4,003.91	0.00	1,930.23	5,934.14	1,781.37
NORTH	4,003.91	7,542.82	4,218.13	100%	100%	100%	4,003.91	7,542.82	4,218.13	15,764.86	1,066.34
NORTH CENTRAL	4,003.91	4,926.87	4,380.20	100%	100%	100%	4,003.91	4,926.87	4,380.20	13,310.98	13,186.74
SOUTH CENTRAL	4,003.91	0.00	1,387.14	100%	100%	100%	4,003.91	0.00	1,387.14	5,391.05	6,622.34
SOUTH	4,003.91	10,064.90	3,498.48	100%	100%	100%	4,003.91	10,064.90	3,498.48	17,567.29	2,950.35
COAST	4,003.91	314.40	7,249.63	100%	100%	100%	4,003.91	314.40	7,249.63	11,567.94	11,499.55

Exhibit 4: Case 4: Future Gen Mix, Ext Load, Limited Availability

Zone	Fut Cap Non	Fut Cap Wind	Fut Cap Solar	Ext Avail Non	Ext Avail Wind	Ext Avail Solar	Total Non	Total Wind	Total Solar	Total Cap	Ext Load
FAR WEST	4,003.91	6,867.58	6,505.29	25.07%	28%	84%	1,003.91	1,922.92	5,464.44	8,391.27	3,936.34
WEST	4,003.91	11,361.10	3,595.21	25.07%	28%	84%	1,003.91	3,181.11	3,019.98	7,204.99	1,768.64
EAST	4,003.91	0.00	1,930.23	25.07%	100%	90%	1,003.91	0.00	1,737.21	2,741.12	2,377.52
NORTH	4,003.91	7,542.82	4,218.13	25.07%	28%	84%	1,003.91	2,111.99	3,543.23	6,659.13	1,210.05
NORTH CENTRAL	4,003.91	4,926.87	4,380.20	25.07%	60%	87%	1,003.91	2,956.12	3,810.77	7,770.81	20,882.51
SOUTH CENTRAL	4,003.91	0.00	1,387.14	25.07%	100%	90%	1,003.91	0.00	1,248.43	2,252.34	10,458.69
SOUTH	4,003.91	10,064.90	3,498.48	25.07%	75%	90%	1,003.91	7,548.68	3,148.63	11,701.22	4,735.78
COAST	4,003.91	314.40	7,249.63	25.07%	75%	90%	1,003.91	235.80	6,524.67	7,764.38	14,647.48

Exhibit 5: Case 5: Current Gen Mix, Ext Load, Added Availability

Zone	Base Cap Non	Base Cap Wind	Base Cap Solar	Add Avail Non	Add Avail Wind	Add Avail Solar	Total Non	Total Wind	Total Solar	Total Cap	Ext Load
FAR WEST	4,003.91	5,143.90	5,270.70	86.00%	86.00%	84.00%	3,443.36	4,423.75	4,427.39	12,294.50	3,936.34
WEST	4,003.91	8,923.10	1,151.60	67.00%	86.00%	84.00%	2,682.62	7,673.87	967.34	11,323.83	1,768.64
EAST	4,003.91	0.00	309.80	86.00%	100.00%	90.00%	3,443.36	0.00	278.82	3,722.18	2,377.52
NORTH	4,003.91	6,266.90	1,343.80	67.00%	86.00%	84.00%	2,682.62	5,389.53	1,128.79	9,200.95	1,210.05
NORTH CENTRAL	4,003.91	2,247.70	540.90	86.00%	86.00%	87.00%	3,443.36	1,933.02	470.58	5,846.97	20,882.51
SOUTH CENTRAL	4,003.91	0.00	346.90	86.00%	100.00%	90.00%	3,443.36	0.00	312.21	3,755.57	10,458.69
SOUTH	4,003.91	7,726.30	608.21	67.00%	86.00%	90.00%	2,682.62	6,644.62	547.39	9,874.63	4,735.78
COAST	4,003.91	151.20	654.00	86.00%	86.00%	90.00%	3,443.36	130.03	588.60	4,161.99	14,647.48

Exhibit 6: Case 6: Future Gen Mix, Ext Load, Added Availability

Zone	Fut Cap Non	Fut Cap Wind	Fut Cap Solar	Add Avail Non	Add Avail Wind	Add Avail Solar	Total Non	Total Wind	Total Solar	Total Cap	Ext Load
FAR WEST	4,003.91	6,867.58	6,505.29	86.00%	86.00%	84.00%	3,443.36	5,906.11	5,464.44	14,813.92	3,936.34
WEST	4,003.91	11,361.10	3,595.21	67.00%	86.00%	84.00%	2,682.62	9,770.55	3,019.98	15,473.14	1,768.64
EAST	4,003.91	0.00	1,930.23	86.00%	100.00%	90.00%	3,443.36	0.00	1,737.21	5,180.57	2,377.52
NORTH	4,003.91	7,542.82	4,218.13	67.00%	86.00%	84.00%	2,682.62	6,486.83	3,543.23	12,712.67	1,210.05
NORTH CENTRAL	4,003.91	4,926.87	4,380.20	86.00%	86.00%	87.00%	3,443.36	4,237.11	3,810.77	11,491.24	20,882.51
SOUTH CENTRAL	4,003.91	0.00	1,387.14	86.00%	100.00%	90.00%	3,443.36	0.00	1,248.43	4,691.79	10,458.69
SOUTH	4,003.91	10,064.90	3,498.48	67.00%	86.00%	90.00%	2,682.62	8,655.81	3,148.63	14,487.07	4,735.78
COAST	4,003.91	314.40	7,249.63	86.00%	86.00%	90.00%	3,443.36	270.38	6,524.67	10,238.41	14,647.48

Exhibit 7: Project Code (.m file)*

See separately submitted.

*Exhibit 8: Project Code (8 *.mat files, one for each case/trial)*

See separately submitted.

Exhibit 9: Project Code Results (.pdf file of code results)*

See separately submitted.

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